

REU 2016 Essential Software

During the Texas A&M REU program this summer, you will be using two software packages: Sage (used for mathematical coding) and LaTeX (used for writing mathematical research papers). We will go over how to use them in detail when you arrive in June. If you plan to bring your own computer, it would help speed things along if you install these software packages before beginning the program.

Below are two sets of installation instructions, one for Mac OS X, and one for Windows. In order to standardize things as much as possible, it would be advisable to update these packages to their latest versions, even if you already have some version of them installed previously. If you run into any problems while installing these packages, feel free to email our graduate assistant Kaitlyn Phillipson at kaitlyn@math.tamu.edu. In particular, if you are running a Linux system and need assistance downloading this software, please send her an email. I look forward to meeting all of you in June!

Mac OS X Instructions

If at any point, your computer says an app cannot be opened due to being from an unidentified developer, try right clicking and selecting open. This should give you a pop up that allows you to open the program.

Latex - Mac OS X

1. Download the installation file at the link below.

`http://mirror.ctan.org/systems/mac/mactex/MacTeX.pkg`

This file is rather large, so I recommended you download it on a relatively fast internet connection.

2. Open the downloaded “MacTeX.pkg” file to launch the LaTeX installer.
3. Follow the onscreen instructions for installation. All of the default settings should be fine. The installation may take a while, but you can feel free to do other things while it installs.
4. Once the installation finishes, you should test your installation. Download and open the following file:

`http://www.personal.psu.edu/dpl14/latex/sample.tex`

It should open in a program called TeXShop (which you just installed). If it opens TeXworks or a different editor, you can close that application, highlight the tex file on Finder, type Cmd-i, go to the “Open with” menu, select TeXshop and click on Change All.

5. Go to the Typeset menu, and select Typeset. This should briefly display some output, and then open a pdf file with the header “Sample L^AT_EX File”. If it does, you are good to go!

Sage - Mac OS X

1. Download the file at the link below.

`http://mirrors.mit.edu/sage/osx/intel/sage-7.2.beta0-OSX\_10.11.4-x86\_64.dmg`

Again, these files are rather large, so you should use a fast internet connection if you have the option. Note: if you choose to download a different installer from the Sage site (www.sagemath.org), it is important you do NOT download the “.app” installer, as it has been known to break certain functionality.

2. Open the .dmg file you downloaded. This will mount what looks like a disk drive on your desktop.
3. Drag the folder “sage” from the DMG file you mounted into the “Applications” folder. This may take some time to copy.
4. Once it has finished copying, right click on the disk drive and select “Eject”.
5. This completes the installation. To test your installation, open the Terminal (located in the Applications folder, then in the Utilities folder). Next, type the following command into Terminal, and hit Enter:

```
/Applications/sage/sage
```

This will launch Sage. It may take up to a minute to open completely the first time while it initializes itself. In the future it will start much quicker. When it finishes opening, you should see the following prompt:

```
sage:
```

6. Test it by running something (relatively) simple, like

```
sage: factorial(100)
```
7. The first time you open a notebook with the command **notebook()**, it will ask you for a password. It is important that you remember this password and even more important that you do not use a trivial password. The notebook will start in your default browser. To finish your session, you do not need to press the logout button. Simply close the browser window and click **Ctrl-C** in the terminal.
8. To exit Sage, enter

```
sage: exit
```

Windows Instructions

Latex - Windows

1. Download the MikTeX installer here:

<http://mirrors.ctan.org/systems/win32/miktex/setup/basic-miktex-2.9.5872-x64.exe>

2. Launch the installer, and follow the onscreen instructions.
3. Once the installation finishes, you should test your installation. Download and open the following file:

<http://www.personal.psu.edu/dpl14/latex/sample.tex>

It should open in a program called Texworks (which you just installed).

4. Go up to the Typeset menu, and select Typeset. As it is compiling, it may prompt you to download additional packages. Just click OK for each dialog. In the end a PDF should open with the header “My Sample L^AT_EX Document”.

Sage - Windows

Unfortunately, installing Sage for Windows is not an easy task. The best way to do so is to emulate Linux using VirtualBox. This page gives very thorough and detailed instructions for installing and running Sage in this manner:

<http://wiki.sagemath.org/SageApplianceInstallation>

Follow these instructions to install and run Sage, and if any issues come up, feel free to send me an email. After Sage is installed you should read the following webpage to learn the basics of using the sage notebook in Windows.

<http://wiki.sagemath.org/SageAppliance>

You can further test that Sage installed properly by running something (relatively) simple, like

```
factorial(100)
```

To exit the Sage notebook follow the instructions in the last webpage linked above.

Using Sage on Windows

Using Sage in your web browser (described below) rather than directly in VirtualBox makes it easier to save and access files.

1. Open VirtualBox and start the Sage virtual machine.
2. Minimize both VirtualBox windows.
3. Open your web browser, and go to the following URL (You may need to wait a couple seconds):

`http://localhost:8000/`

If that URL does not load, try the following instead:

`http://localhost:8888/`

4. From here, you can use Sage with Username: *admin* , Password: *sage*